

Step by step instructions to implement resumes-ai-rag

01: Create a new folder 'resumes-ai-rag'

02: Add folder to workspace

03: Add Architecture.md file to this folder

04: Open co-pilot chat and choose 'Ask' mode and 'GPT-5-mini' model 05:

Use this template to start :

'I am building Resume Search Algorithm using AI - RAG approach and the architecture document is attached to the context. Start with creating prompt file(s), instructions that can be used by co-pilot side chat for generating / editing / debugging code for the project'

06: Drag and drop 'Architecture.md' to copilot and paste the above content and click execute.

07: If you are good with the prompt files, change to 'Agent' mode and ask to generate:

'I am okay with prompt files and instructions - create them now for my usage. Follow enterprise grade solutions'

08: You will notice the base prompt and templates are created

09. Next is to generate the code - start with agent mode

Generate Project scaffold + health endpoints + MongoDB connection.

10. You will notice the folders and files are generated but may show errors

11. Open the terminal and run 'npm install' that will resolve the errors

Note: There will be errors due to non-loaded components

(package.json : mongo version : 5.7.0, if error comes like : No matching version found)

12. Click Keep button on sidebar to store them locally

13. If you still see errors — ask agent: "Fix the errors in workspace", then drag & drop those error files alone (agent will fix missing dependencies/compatibility). **Confirm errors are fixed.**

14. copy .env.example to .env file

15. Update your mongodb connection(db_url, db_name, collection_name, bm25_index, vector_index) in the .env and save it .

16. Now in the terminal - do

'npm run dev'

17. Open the postman and hit healthcheck endpoint <http://localhost:3000/v1/health>

18. If it is failing, ask agent to fix it by providing the message / error that you notice in postman or terminal.

19. Continue with mongodb health check <http://localhost:3000/v1/health/db>
20. Change to Ask Mode. Drag and drop “Architecture.md” and “.env” file. 'To implement Vector Embedding + `/v1/embeddings` with Architecture.md and create_endpoint.md prompt file. Confirm if you have everything to code (Do not generate code)>> (drag + drop)
21. Confirm references are used - note if any questions: - answer accordingly (avoid batches, avoid hardcodings)
22. Switch to Agent mode >> Create the mistral model name and api key in the '.env' file before you proceed
- 23 Update manually >> .env file with mistral key and dimension as 1024 and save it

```
EMBEDDING_MODEL=mistral-embed  
EMBEDDING_DIM=1024
```
24. Generate code 'To implement Vector Embedding + `/v1/embeddings` with Architecture.md and create_endpoint.md prompt file' with mistral-embed for embeddings (store in .env) as single input at time.
- 25 If agent updates package.json and ask you to approve 'npm install' accept those.
- 26 stop the running server (ctrl + c) in the terminal and run 'npm run dev' again
- 27 go to postman and confirm <http://localhost:3000/v1/embeddings>

```
{"input": "hello world"}
```
- 28 If all good, continue to next step - else go to debugging
- 29 Switch to Agent mode >> Create the groq LLM model name and api key in the '.env' file before you proceed
- 30 Update manually >> .env file with groq key and model name and save it

```
GROQ_API_KEY=your_groq_api_key_here  
GROQ_LLM_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
```
- 31 Click 'Keep' and Restart the server terminal
- 32 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'Atlas Search BM25 config + `/v1/search/bm25`.'
- 33 Confirm service, routes and are created and index is updated
- 34 Restart the server and test the following ‘Debugging Notes’
- 35 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'Vector Search config + `/v1/search/vector`'. DON'T use knn beta.
- 36 Confirm service, routes and are created and index is updated.

- 37 Restart the server and test the following 'Debugging Notes'
- 38 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'Hybrid Search(BM25 and vector) config + `/v1/search/hybrid`.'
- 39 Confirm service, routes and are created and index is updated.
- 40 Restart the server and test the following 'Debugging Notes'
- 41 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'LLM Re-Ranking config + `/v1/search/rerank`.'
- 42 Confirm service, routes and are created and index is updated
- 43 Restart the server and test the following 'Debugging Notes'
- 44 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'Summarization config + `/v1/search/summarize`.'
- 45 Confirm service, routes and are created and index is updated
- 46 Restart the server and test the following 'Debugging Notes'
- 47 Agent Mode :: Drag and drop 'Architecture.md' to co-pilot. Implement 'End-to-End Resume Search Pipeline config + `/v1/search`.'
- 48 Confirm service, routes and are created and index is updated
- 49 Restart the server and test the following 'Debugging Notes'

Debugging Notes:

- If there are issues - follow these steps:

- 1) Read the error messages (terminal, postman)
- 2) Change to Ask mode - 'Analyse the error and provide crisp updates' + error
- 3) Read the fix and if you are comfortable - fix it !